Mapping & Analyzing Participation

Suggestions and Guidance for Future Research Teams

E483/E583 Information Visualization — Spring 2026
Indiana University Luddy School of Informatics, Computing, and Engineering

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Contents

Project Overview

This document is written for researchers and development teams who continue work on the **Mapping and Analyzing Participation** project after the Spring 2026 semester. The project was commissioned by Daniel F. Bassill of the **Tutor/Mentor Connection** (T/MC, 1993–2011) and its successor **Tutor/Mentor Institute LLC** (TMI, 2011–present) to analyze and visualize attendance records from 42 Tutor/Mentor Leadership and Networking Conferences held in Chicago between May 1994 and November 2015. When referencing the client organization, use T/MC for events prior to 2011 and Tutor/Mentor Institute LLC for anything after; Bassill uses both names throughout his published materials.

The primary dataset is:

`ProjectFiles/All_Conferences_Data_Cleaned-2026 - Sheet1.csv`

It contains 6,410 participation records across 18 columns, covering roughly 2,161 unique organizations and 21 years of network-building activity in Chicago's high-poverty neighborhoods.

The codebase is a **Django** web application with a `Pandas/ Plotly/NetworkX` analytics layer. All source code lives in the `conferences/` app. Docker support is included via `Dockerfile` and `docker-compose.yml`.

Understanding the Data Before You Touch It

The SNA Category System

Every attendance record carries an `sna_category` field drawn from a controlled vocabulary of eleven canonical types:

Category	What it represents
Program	Tutoring/mentoring program providers
Business	Corporate sponsors and private-sector partners
College	Universities and community colleges
K-12 School	Public and private K-12 institutions
Government	Municipal, county, state, and federal agencies
Faith	Faith-based organizations
Foundation	Grant-making foundations and philanthropies
Intermediary	Capacity-building and coalition organizations
T/MC	Tutor/Mentor Connection itself
Resource	Consultants and support services
Media	Journalists, reporters, and media organizations
Other	Records that could not be classified

The original column (`sna_category_original`) contained hundreds of inconsistent spellings. The normalization mapping lives in `conferences/sna_normalize.py`. **Consult Daniel**

Bassill before adding or reclassifying categories—the vocabulary is intentional and affects downstream network analysis.

Bassill has explicitly requested a **12th category: Media** (journalists and media organizations). This category is intentionally thin in the current dataset, and that sparsity is itself meaningful—building media coverage is a core conference goal, and its absence in the data reflects a real gap in outreach.

Organization Name Cleaning

The `organization` field (cleaned) is the authoritative identifier for network analysis. The `organization_original` field preserves the raw value as entered at each conference. Never use the original field as a join key—variant spellings of the same organization appear across years.

The cleaning was performed manually by the client in the 2026 CSV revision. If the dataset is extended with new records, the same cleaning step must be applied before loading data into Django.

Known Data Quality Issues

- A small number of records have a blank `organization` field. These are excluded from all network exports and most analytics.
- Records classified as `Other` may be reclassifiable with additional domain knowledge from the client.
- The `active_inactive` column is populated inconsistently and was not used in the Spring 2026 analysis. It may carry useful longitudinal signal for organizations that later closed.
- Address and ZIP code fields are sparsely populated for earlier years (pre-2000). Geographic analysis is more reliable for 2000 onward.
- **Organization names change over time.** An organization that attended in the 1990s may appear under a different name in later years—sometimes a third name by the 2010s. The cleaned `organization` field represents the name as used at that conference; cross-decade queries may miss the same entity under a variant name.
- **Individuals moved between organizations.** A person who attended four conferences through three different employers will appear as three separate organizations each with one attendee, not as one person with four appearances. The first/last name fields can be used to surface this pattern without exposing names publicly.

Technical Infrastructure

Running the Application

With Docker (recommended):

```
docker-compose up --build
```

The app will be available at `http://localhost:8000`.

Without Docker:

```
pip install -r requirements.txt
python manage.py migrate
python manage.py load_data   # loads the CSV into SQLite
python manage.py runserver
```

Loading Data

The management command `conferences/management/commands/load_data.py` reads the CSV, applies SNA normalization (via `sna_normalize.py`), and populates the `Attendance` table. Re-running it with a revised CSV will not duplicate records because it clears and reloads the table. If you extend the dataset, verify the column headers match those expected by the loader before running.

Database

The application ships with SQLite (`db.sqlite3`). For a production deployment with concurrent users, migrate to PostgreSQL. The Django ORM abstracts this entirely—update `DATABASES` in `MappingAndAnalyzingParticipation/settings.py` and add `psycopg2-binary` to `requirements.txt`.

Environment Variables

Sensitive settings are stored in `.env` (not committed to git). A `SECRET_KEY` must be set before deployment. See `.env` for the variable names expected.

Network Analysis Guidance

Available Exports

The application exposes four CSV export endpoints under `/export/`:

Endpoint	Contents
`/export/nodes/`	One row per organization with sector, event count, and record count
`/export/edges/org-event/`	Bipartite edges: organization ↔ conference event
`/export/edges/org-org/`	Co-attendance edges with shared-conference weights
`/export/edges/event-sector-org/`	Three-layer (event → sector → org) edge list

All endpoints accept `?year=`, `?sector=`, and `?state=` query parameters for filtered exports.

Importing into Kumu.io

The application exposes JSON endpoints (e.g. `/export/edges/org-event/`) that Kumu can link to directly. This is the preferred approach because the map updates automatically when the underlying data changes:

1. Copy the JSON export URL from the Network Maps page on the running app.
2. In Kumu, choose **Import → Link map to remote JSON** and paste the URL.
3. Use **Import → Re-import** to force a refresh without recreating the map from scratch.

Alternatively, for a fully static import via CSV:

1. Download `nodes.csv` and `edges_org_org.csv`.
2. In Kumu, create a new map and use **Import → CSV**.
3. Map `Id/Label` to Kumu's element fields and `SNA_Category` as the element type for color coding.
4. Use `Event_Count` as the size field to weight nodes by participation history.

When embedding network maps in the Django app, **link to the live Kumu map URL** rather than rendering a static PNG. A linked map lets visitors open a full interactive view, which is significantly easier to explore than a small embedded image.

Importing into Gephi

1. Download `nodes.csv` and `edges_org_org.csv`.
2. In Gephi, use **File → Import Spreadsheet**. Import nodes first, then edges.
3. Set `Source/Target` as edge columns and `Weight` as the edge weight.
4. Run **ForceAtlas 2** layout. Set node size to `Event_Count` and color by `SNA_Category`.
5. Recommended analysis modules: modularity (community detection), betweenness centrality, and eigenvector centrality to identify bridge organizations.

Suggested Network Research Questions

- Which organizations served as consistent *bridges* between sectors (high betweenness centrality) across the full 21-year span?
- Did the network become more or less cross-sector connected over time? Run year-sliced graphs and compare modularity scores.
- Are there organizations that attended many early conferences but disappeared from later ones? Do they correlate with the `active_inactive` field?
- Which sector pairs co-attended most frequently? This can inform partnership recommendations for T/MC's current outreach.

Recommended Next Steps

High Priority

1. **Geographic mapping.** ZIP codes are present for a substantial portion of records. Joining to a Chicago neighborhood shapefile (available from the Chicago Data Portal) would allow choropleth maps showing which neighborhoods were most represented and which were underserved—directly relevant to Bassill's outreach mission.
2. **Organization URL enrichment completion.** The script `find_org_websites.py` located URLs for most organizations and saved results to `org_websites.csv`. These URLs have not yet been joined to the Django database. Adding a `website` field to the `Attendance` model and linking via `organization` name would enable clickable org profiles in the dashboard.
3. **Manual review of `unverified` and `not found` URL results.** Roughly a quarter of organizations in `org_websites.csv` have status `unverified` or `not found`. Cross-referencing with the Tutor/Mentor Exchange directory at tutormentorexchange.net would improve coverage. Note: always use the `.net` domain— `.com` does not resolve.
4. **OHATS data integration.** The OHATS (Organization Health Assessment Tool & Survey) dataset is partially cleaned and ready for a future team to pick up. It is a strong candidate for the next semester project because it is data-visualization-heavy and connects directly to Bassill's four strategic pillars (see the "Strategy" concept map on tutormentorexchange.net under Tutor/Mentor Connection → Strategy). Future teams should read the OHATS section of the project handbook before beginning.
5. **Add a "Media" SNA category.** Update `conferences/sna_normalize.py` to map media-related original values to a new `Media` category and remove `Resource` entries that properly belong there. Confirm the mapping with Bassill before committing.

Medium Priority

1. **Longitudinal trend analysis.** The participation frequency chart currently shows absolute attendance counts. Normalizing by the number of organizations *known to exist* in each year (using external directories) would convert raw counts to participation rates—a more meaningful metric for researchers.
2. **Individual-level analysis.** The dataset includes first and last names, enabling tracking of individual attendees across conferences. Identifying individuals who attended the most events, changed organizations over time, or bridged sectors as individuals (not just organizationally) would add a person-centered perspective to the network view. Be mindful of privacy: individual-level data should not be published without Bassill's explicit approval.
3. **Interactive network visualization in-browser.** The three current network maps are static PNGs rendered server-side by Matplotlib. Replacing them with a D3.js or Sigma.js force-directed graph would allow users to hover, filter, and explore nodes interactively

without downloading Gephi or Kumu.

4. **Dashboard usage guide.** The category/year/conference filter controls are not self-explanatory to a first-time visitor. Adding a brief on-page explanation or tooltip layer describing what each filter does—and how the top-10 view can surface duplicate or mis-categorized organizations—would significantly improve usability.
5. **Link to fall 2025 second group's work.** A second team also analyzed the conference data in fall 2025 (in addition to Team K). Both teams' work should be linked under the "More" dropdown in the Network Maps section of the app so the full lineage of this project is visible to future researchers.
6. **Cross-reference the printed conference directories.** Bassill produced printed directories of invited organizations from 1994 to 2003 (available as PDFs). Comparing those invitation lists to the actual attendance records would reveal who was invited but never attended—a meaningful gap for outreach analysis.
7. **Public deployment.** The Docker configuration is ready. Deploying to a cloud provider (Render, Railway, or a university-hosted server) with a persistent PostgreSQL database would make the tool accessible to Bassill and the broader research community without requiring a local Python environment.

Low Priority / Long-Term

1. **Comparative analysis.** The network structure of this Chicago conference series could be compared to similar civic networking efforts in other cities using published SNA datasets, positioning this work within the broader nonprofit network literature.
2. **IU School of Philanthropy collaboration.** Brad Fulton (IU Lilly Family School of Philanthropy) leads Project 990, which maps nonprofit organizations using IRS data. Connecting that geographic nonprofit database to the conference participation data could reveal which types of organizations are most underrepresented at conferences relative to their presence in a given community. Fulton has connected with Bassill on LinkedIn and has expressed interest; reaching out through the IU Data Science Club or an independent study arrangement is a natural entry point.
3. **Donor pipeline layer.** Bassill described a concept for aggregating donor-flow data on top of the geographic nonprofit map: tracking how many visitors arrived at a nonprofit's donation portal through a shared platform, broken down by ZIP code and time period. This is a longer-term product idea that would require partnerships with nonprofit payment processors and is well suited to a capstone or MBA-level project team.
4. **Visualization competition outreach.** The IU Luddy hackathon and Tableau's annual Iron Viz competition are both viable venues for surfacing this dataset to a wider student audience. Bassill has also pointed to existing websites that aggregate student visualization work as models for a permanent archive of conference-data visualizations done by successive teams.

Working With the Client

Daniel F. Bassill (`tutormentor1@gmail.com`) is the domain expert for all questions about the data's meaning, the SNA category definitions, and organizational context. He has deep knowledge of Chicago's tutoring and mentoring ecosystem that no documentation can fully substitute.

When writing about the client organization, note that **Tutor/Mentor Connection** (T/MC) is the historically correct name for the 1993–2011 era; **Tutor/Mentor Institute LLC** (TMI) is the current legal entity (2011–present). Bassill uses both names in his published materials and PDF essays. Do not refer to T/MC as the current organization—doing so understates the institutional transition. Blog resources are at tutormentor.blogspot.com; tags for "network analysis," "Kumu," and "tipping point" are particularly useful starting points for new team members.

Key things to confirm with him before major analytical decisions:

- Any reclassification of `Other` records into named categories.
- Any changes to organization name normalization.
- Whether individual-level data (names) can appear in public-facing views.
- The intended audience for any new visualizations—his four stakeholder groups (T/MC, nonprofits, researchers, general public) have different technical literacy levels.

Repository and File Reference

Path	Purpose
`conferences/models.py`	Django ORM model for attendance records
`conferences/views.py`	All dashboard, chart, export, and map views
`conferences/sna_normalize.py`	SNA category normalization mapping
`conferences/management/commands/load_data.py`	CSV ingestion command
`ProjectFiles/*.csv`	Source data and org URL output
`find_org_websites.py`	Web crawler for organization URLs
`generate_network_maps.py`	Standalone script for batch PNG export
`templates/conferences/`	HTML templates for all views
`docker-compose.yml`	Container orchestration for local/prod use

Use of AI-Assisted Development

The Spring 2026 team developed the entire codebase using **Claude Code** (Anthropic, `claude-sonnet-4-6`) as a development tool. This includes some views and exports, the organization URL crawler, and this document.

Future teams should feel free to continue this approach. Claude Code can read the existing codebase and extend it, provided the developer can articulate the goal clearly. The `.tex` writeups in `ProjectFiles/` document methodology and prompts used, and can serve as a model for documenting AI-assisted work in future research outputs.